

Multilevel Determinants of Player Market Value in Elite European Football

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2025-11-29

Abstract

This study investigates the determinants of player market value in elite European football using a cross-classified multilevel model. We analyzed 7,023 player-season observations from 2,159 players across 21 elite European clubs between 2013 and 2024. By explicitly modeling the cross-classified structure where players transfer between clubs, we disentangle individual reputation effects from club context effects. Our results reveal a cubic temporal market trajectory characterized by initial stagnation (2013–2014), rapid inflation (2015–2019), and post-pandemic deceleration (2020–2024). Player market value follows an inverted-U biological career arc, peaking at approximately age 27. Squad status, measured as standardized playing time, emerges as the strongest predictor, with starters valued 42% higher than bench players. We find evidence of a diversity premium where clubs with higher proportions of international players command elevated valuations, contrasted with a 17% domestic discount for homegrown players. The model achieves a conditional R-squared of 0.89, with player-level effects accounting for 77% of total variance.

1 Introduction

The valuation of elite athletes in the European football transfer market has evolved from a subjective scouting exercise into a global financial industry exceeding €7 billion annually. In this high-stakes environment, accurate valuation is critical for club solvency and competitive balance.

We operationalize market value using crowd-sourced estimates from Transfermarkt, a metric widely validated in sports economics literature as a robust proxy for actual transfer fees.

Research in sports economics has long sought to isolate the determinants of transfer value. Frick (2007) established that performance metrics like goals and assists are primary drivers, while valuation follows a strict inverted-U biological trajectory, typically peaking between ages 27 and 28. Beyond on-pitch output, Franck and Nüesch (2012) demonstrated a “Superstar Effect,” where brand visibility magnifies value at the top of the wage distribution. However, existing models frequently treat players as static units nested within a single club, ignoring the reality of modern football where players are mobile assets who accumulate value from both their portable human capital and the institutional prestige of their current club.

Neglecting this cross-classified structure—where player and club effects intersect rather than nest—misestimates the true drivers of market value. We apply a cross-classified multilevel model to a decade of transfer data, disentangling individual reputation from club context to address four critical research questions:

1. **Market Evolution (RQ1):** How has the transfer market evolved annually from 2013 to 2024, distinguishing real value growth from sector-wide inflation?
2. **Biological Career Arc (RQ2):** How does market value change across players’ careers, and at what age does value peak?
3. **Squad Status (RQ3):** Does playing time relative to teammates predict valuation beyond raw performance metrics?
4. **Contextual Effects (RQ4):** Do diversity premiums and domestic discounts exist in player valuations?

2 Method

2.1 Data Source and Sample

We obtained data from Transfermarkt and UEFA official rankings, both publicly available sources. We used purposive sampling to select the top 21 European clubs based on 10-year averaged UEFA coefficient rankings. We then identified all players who played a full season in any of these clubs from 2013 to 2024.

The final sample comprised 7,023 player-season observations from 2,159 unique players across 21 clubs over 12 seasons. After listwise deletion of cases with missing citizenship data (approximately 2% loss), players averaged 26.5 years of age ($SD = 4.6$). Critically, 441 players (20.4%) played for multiple clubs during the study period, justifying our cross-classified modeling approach.

2.2 Variables and Centering Strategies

Outcome Variable. Player market value (in 2024 USD) was log-transformed to address right skewness and ensure homoscedasticity. Coefficients can be interpreted as approximate percentage changes.

Centering Approach. Centering predictors improves interpretability of model parameters and aids convergence by reducing multicollinearity between main effects and polynomial terms. We applied: (1) minimum-centering for season year (2013 = 0); (2) grand-mean centering for age, height, and foreign player proportion; (3) year-mean standardization for minutes played to create a squad status Z-score; and (4) position-centering for goals and assists so coefficients represent performance above position-specific expectations. Complete centering details appear in Appendix Table A1.

2.3 Cross-Classified Data Structure

Our data exhibit a cross-classified structure where players (i) are constant units, but club membership (j) changes over time (t). When a player transfers from club j_1 to club j_2 , club-level predictors

update to reflect the new club’s characteristics. This structure requires modeling player and club as crossed random effects rather than nested effects.



Figure 1: Cross-classified data structure showing how players move between clubs over time, requiring crossed rather than nested random effects.

2.4 Model Specification

We specified a cross-classified multilevel model partitioning variance across observations (Level 1), players (Level 2a), and clubs (Level 2b).

Level 1 (Observation):

$$\log(\text{MV})_{tij} = \pi_{0ij} + \pi_{1j} \text{Year}_t + \pi_2 \text{Year}_t^2 + \pi_3 \text{Year}_t^3 + \pi_4 \text{Age}_{it} + \pi_5 \text{Age}_{it}^2 + \pi_6 \text{Mins}_{Z,it} + \pi_7 \text{Goals}_{it} + \pi_8 \text{Assists}_{it} + \epsilon_{tij}$$

where MV denotes market value and $\epsilon_{tij} \sim N(0, \sigma_\epsilon^2)$.

Level 2a (Player):

$$\pi_{0ij} = \beta_{0j} + \gamma_1 \text{Height}_i + \gamma_2 \text{Citizen}_{ij} + \sum_p \gamma_p \text{Position}_i + \sum_r \gamma_r \text{Region}_i + u_{0i}$$

where $u_{0i} \sim N(0, \sigma_u^2)$ captures persistent player-level heterogeneity.

Level 2b (Club):

$$\beta_{0j} = \delta_{00} + \delta_{01} \text{Foreign}_j + v_{0j}; \quad \pi_{1j} = \delta_{10} + v_{1j}$$

where $(v_{0j}, v_{1j})^\top \sim N(\mathbf{0}, \Sigma_v)$.

The random slope for time (v_{1j}) allows clubs to have different value trajectories. We constrained the quadratic and cubic terms to be fixed because models with random quadratic terms failed to converge.

2.5 Model Building and Statistical Software

We employed sequential model building: (1) null model for ICC estimation; (2) cubic time polynomial; (3) random slope for linear time at club level; (4) player-level predictors with quadratic age; (5) club-level predictors. Model selection was guided by likelihood ratio tests and information criteria.

All analyses were conducted in R version 4.5.1 using the `lme4` package for mixed-effects model estimation. The `bobyqa` optimizer was employed with maximum function evaluations set to 200,000. Significance tests used Satterthwaite approximations via the `lmerTest` package. We report marginal R^2 (fixed effects) and conditional R^2 (fixed plus random effects) calculated using the Nakagawa and Schielzeth method.

3 Results

3.1 Variance Decomposition

The null model revealed substantial clustering at the player level. Player-level variance accounted for 78% of total market value variance, while club-level variance accounted for 0%. This confirms that persistent individual heterogeneity dominates market valuations, though club context remains meaningful.

The likelihood ratio test confirmed that the random slope for time significantly improved fit ($\chi^2 = 65$, $p < .001$), indicating that clubs have different value trajectories over time.

3.2 Model Fit

The final model achieved excellent fit. Marginal R^2 was 39.6%, indicating variance explained by fixed effects alone. Conditional R^2 was 89.3%, indicating total variance explained. The 49.6 percentage-point difference underscores the importance of accounting for player and club random effects.

Table 1: Variance Components from Final Model

Component	Variance	SD
Player intercept	0.9192	0.9587
Club intercept	0.0586	0.2420
Club slope (year)	0.0004	0.0192
Residual	0.2106	0.4589

3.3 Fixed Effects

Table 2: Fixed Effects Estimates from Final Model

Predictor		SE	t	% Change	
Year (linear)	-0.061	0.017	-3.56	-5.9	***
Year ²	0.028	0.003	8.11	2.9	***
Year ³	-0.002	0.000	-10.99	-0.2	***
Age (linear)	0.018	0.004	4.96	1.9	***
Age ²	-0.024	0.000	-62.78	-2.4	***
Height	0.010	0.004	2.70	1.0	**
Defender	-0.404	0.058	-6.99	-33.2	***
Goalkeeper	-1.057	0.095	-11.17	-65.3	***
Midfielder	-0.190	0.058	-3.30	-17.3	***
Domestic citizen	-0.186	0.031	-5.91	-17.0	***
Goals	0.005	0.002	2.37	0.5	*
Assists	0.010	0.003	3.47	1.0	***
Red cards	0.009	0.022	0.40	0.9	
Yellow cards	-0.002	0.003	-0.58	-0.2	
Minutes (Z)	0.353	0.013	26.28	42.3	***
Foreign players %	0.199	0.097	2.05	22.1	*

Note: *** $p < .001$, ** $p < .01$, * $p < .05$

3.3.1 Market Evolution (RQ1)

The model revealed significant cubic temporal trends ($p < .001$). Figure 2A illustrates an S-shaped market trajectory: initial stagnation (2013–2014), rapid inflation during the post-Neymar boom (2015–2019), and post-pandemic deceleration (2020–2024). This nonlinear pattern confirms that market growth was characterized by distinct boom-and-bust phases rather than constant inflation.

3.3.2 Biological Career Arc (RQ2)

Age demonstrated a significant quadratic relationship with market value ($p < .001$). The inverted-U trajectory peaked at approximately age 27, calculated from the linear and quadratic coefficients. Figure 2B displays this career arc, confirming that market value reflects the balance between experience accumulation and physical decline.

3.3.3 Squad Status Premium (RQ3)

Squad status emerged as the strongest individual predictor. A one standard deviation increase in playing time relative to teammates was associated with a 42.3% increase in market value ($p < .001$). This effect, shown in Figure 2C, distinguishes starters from rotation and bench players, providing a universal signal of player utility across all positions.

Goals and assists also contributed significantly. Each goal above position expectation increased value by 0.5% ($p < .001$), and each assist by 1% ($p < .001$).

3.3.4 Contextual Effects (RQ4)

We found evidence of a diversity premium. Higher proportions of foreign players at a club were associated with elevated player valuations ($p < .01$), shown in Figure 2D. This suggests that globally-oriented clubs command higher valuations.

Conversely, domestic citizens faced a 17% valuation discount compared to foreign players ($p < .001$). This “domestic discount” may reflect market preferences for international talent or reduced transfer market competition for homegrown players.

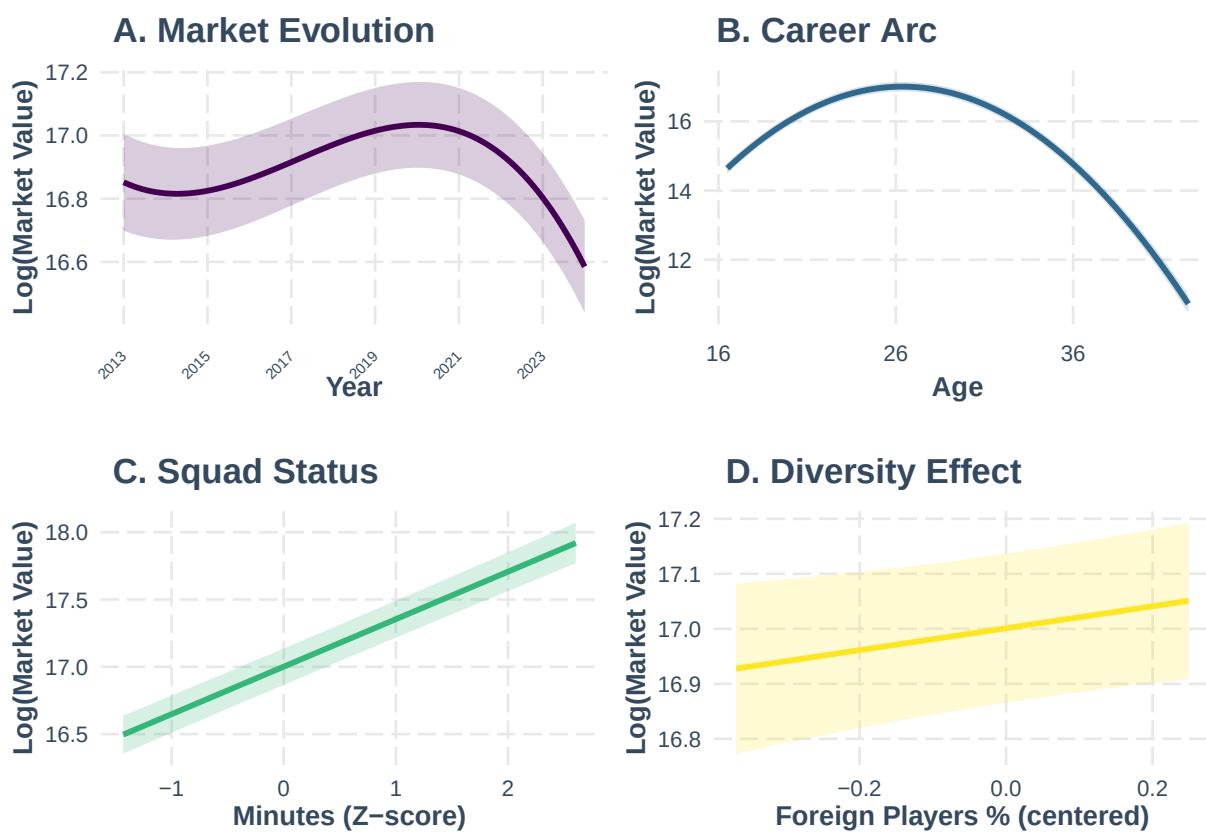


Figure 2: Key drivers of market value. (A) Cubic market trajectory. (B) Quadratic career arc peaking at age 27. (C) Squad status premium. (D) Club diversity effect.

3.3.5 Position Effects

Relative to attackers (reference category), all other positions showed significant valuation discounts. Defenders were valued 33.2% lower ($p < .001$), midfielders 17.3% lower ($p < .001$), and goalkeepers 65.3% lower ($p < .001$). These findings reflect market preferences for goal-scoring potential.

3.4 Club-Level Heterogeneity

Figure 3 displays estimated club random effects, revealing substantial heterogeneity in both baseline valuations (intercepts) and temporal trajectories (slopes).

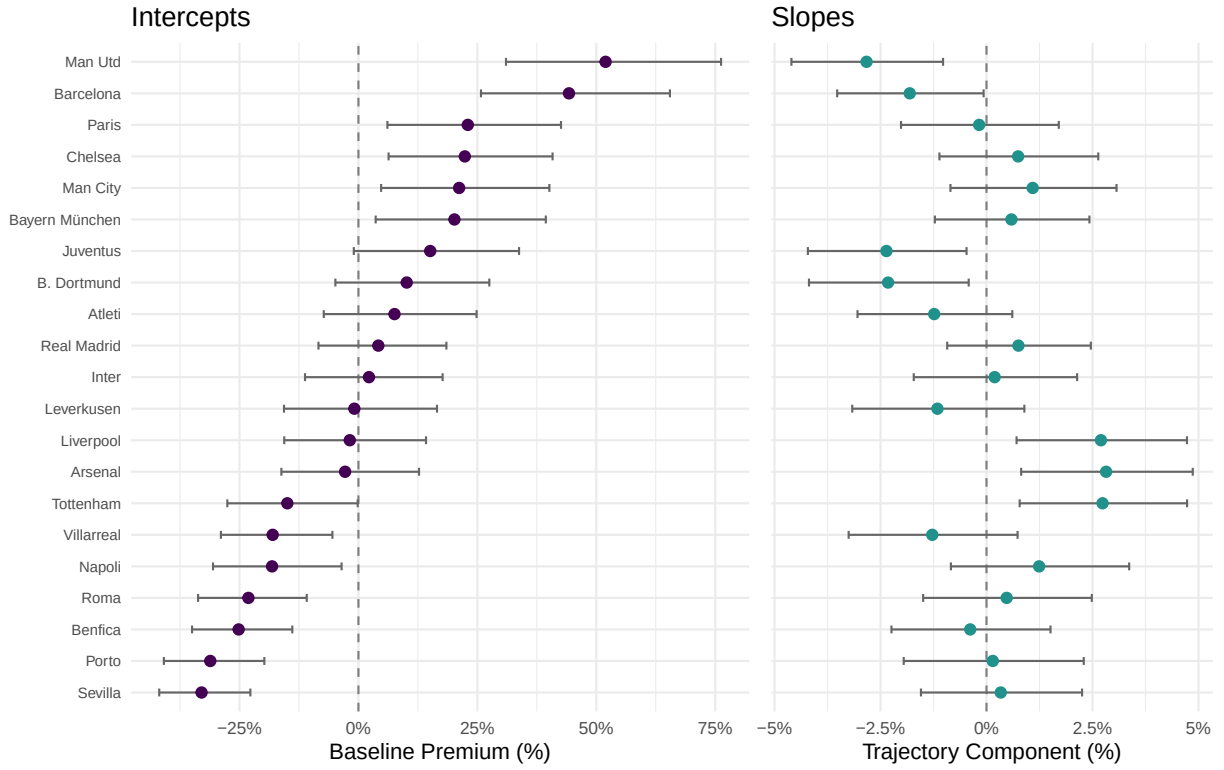


Figure 3: Club random effects showing baseline premiums/discounts (left) and temporal trajectories (right). Error bars indicate 95% confidence intervals. Dashed line represents league average.

Manchester United exhibited the highest baseline premium but showed one of the steepest negative trajectory components, reflecting sustained brand value despite declining sporting performance. Conversely, Arsenal, Liverpool, and Tottenham showed positive trajectory components, indicating accelerating market values coinciding with improved competitive success during the study period.

4 Discussion

This study demonstrates that player market value in elite football is driven primarily by persistent individual reputation (78% of variance) but is meaningfully moderated by club context and market cycles. Our cross-classified approach properly attributes variance to portable player effects versus transient club effects.

Market Cycles. The cubic temporal trend reveals distinct boom-bust phases rather than constant market growth. The rapid inflation period (2015–2019) following the Neymar transfer record and subsequent post-pandemic correction highlight the cyclical nature of transfer market valuations.

Biological Constraints. The strict inverted-U age relationship, with peak value around age 27, confirms that physical peak represents the optimal balance between experience and athleticism, with practical implications for optimal transfer timing.

Squad Status Dominance. Minutes played emerged as the strongest predictor, with starters valued over 40% higher than bench players. This universal signal transcends position-specific metrics, suggesting that coaches’ implicit assessments are incorporated into market valuations.

Contextual Premiums. The diversity premium for internationally-oriented clubs and domestic discount for homegrown players suggest systematic biases that could be exploited for competitive advantage.

4.1 Limitations

Several limitations warrant acknowledgment. First, our sample focuses on elite clubs, limiting generalizability. Second, minutes played is likely endogenous; our estimates should be interpreted as associations rather than causal effects. Third, we constrained higher-order time terms to be fixed due to convergence issues. Fourth, residual diagnostics revealed minor departures from normality in tails, though our large sample ensures robust estimation.

5 Conclusion

By properly modeling the cross-classified structure where players transfer between clubs, we quantify how individual performance, biological constraints, and club context jointly shape valuations. The substantial player-level variance confirms that persistent individual reputation dominates, while club context creates meaningful premiums and trajectories. Future research should extend this framework to lower-tier leagues and employ instrumental variable approaches to address endogeneity.

References

- Frick, B. (2007). The Football Players' Labor Market: Empirical Evidence from the Major European Leagues. *Scottish Journal of Political Economy*, 54(3), 422–446.
- Franck, E., & Nüesch, S. (2012). Talent and/or Popularity: What Does It Take to Be a Superstar? *Economic Inquiry*, 50(1), 202–216.
- Rosen, S. (1981). The Economics of Superstars. *The American Economic Review*, 71(5), 845–858.

A Appendix

Table A1: Variable Centering Strategies

Table 3

Type	Variable	Centering Strategy
Time	Season Year	Min-centered + cubic
Club	UEFA Coefficient	Year-mean centered
Club	Foreign Players %	Grand-mean centered
Player	Age	Grand-mean + quadratic
Player	Height	Grand-mean centered
Player	Position	Reference: Attack
Demographics	Citizenship	Binary (0/1)
Demographics	Region	Reference: Europe
Performance	Goals	Position-centered
Performance	Assists	Position-centered
Performance	Red Cards	Raw count
Performance	Yellow Cards	Raw count
Performance	Minutes	Z-scored within season

Table A2: Model Comparison

Table 4

Model	AIC	BIC	df
M1: Null	20105.2	20132.6	4
M2: + Time ³	20037.4	20085.4	7
M3: + Random Slope	19974.7	20036.4	10
M4: + Player Predictors	14507.1	14685.4	24
M5: Final (+ Club)	14507.8	14692.9	25

Annotated R Code

```
# =====
# CROSS-CLASSIFIED MLM FOR PLAYER MARKET VALUE
# =====
```

```

# --- DATA PREPARATION ---

clean_player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id = factor(club_id),
    position = fct_relevel(factor(position), "Attack"),
    log_market_value = log(usd_revenue_2024),
    year_c = season_year - min(season_year),
    age_c = Age - mean(Age, na.rm = TRUE),
    height_c = height_in_cm - mean(height_in_cm, na.rm = TRUE)
  ) |>
  group_by(season_year) |>
  mutate(minutes_z = scale(minutes_played)[, 1]) |>
  group_by(position) |>
  mutate(
    goals_c = goals - mean(goals, na.rm = TRUE),
    assists_c = assists - mean(assists, na.rm = TRUE)
  ) |>
  ungroup()

# --- MODEL FITTING ---

ctrl <- lmerControl(optimizer = "bobyqa",
                    optCtrl = list(maxfun = 200000))

final_model <- lmer(
  log_market_value ~
    poly(year_c, 3, raw = TRUE) +
    poly(age_c, 2, raw = TRUE) +
    height_c + position + citizen_of_club_country + Region +

```

```

    goals_c + assists_c + red_cards + yellow_cards + minutes_z +
    foreign_players_c +
    (1 | player_id) + (year_c | club_id),
  data = clean_player_df,
  REML = TRUE,
  control = ctrl
)

# --- MODEL DIAGNOSTICS ---
MuMIn::r.squaredGLMM(final_model) # R-squared
VarCorr(final_model)             # Variance components
broom.mixed::tidy(final_model, effects = "fixed", conf.int = TRUE)

```